CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

College Name: A.V.C. College of Engineering

Project Title: Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce

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Project Overview

Customer support teams receive hundreds of support requests every day. Manually reviewing each ticket, identifying its urgency, and assigning it to the correct support agent is time-consuming and can delay the resolution of critical customer issues.

This project provides an intelligent solution using **Salesforce Agentforce** and **Salesforce Flow Automation** to automatically analyze support ticket descriptions, predict their priority, assign them to the appropriate support agent, and create urgent tasks when necessary.

The system reduces manual effort, improves customer satisfaction, increases team productivity, and ensures that critical issues are handled immediately.

Problem Statement

Many organizations still depend on manual ticket prioritization and assignment. This causes:

- Slow response time
- Incorrect ticket assignment
- Increased workload for support teams
- Delayed resolution of urgent issues
- Poor customer satisfaction

This project solves these problems by using AI-based automation through Salesforce Agentforce.

Objectives

- Automatically classify support tickets.
- Predict ticket priority.
- Assign tickets to the correct support agent.
- Reduce manual work.
- Improve response time.
- Increase customer satisfaction.
- Automate task creation for urgent issues.
- Improve overall support team efficiency.

Key Features

- Automatic Ticket Priority Prediction
- AI-Based Ticket Analysis
- Keyword-Based Priority Detection
- Automatic Agent Assignment
- Auto Task Creation for High Priority Tickets
- Salesforce Flow Automation
- Agentforce Integration
- SLA Risk Identification
- Faster Customer Support Process
- Improved Productivity

Technologies Used

- Salesforce CRM
- Salesforce Agentforce
- Salesforce Flow Builder
- Custom Objects
- Lookup Relationships
- Decision Elements
- Assignment Elements
- Record Creation
- Automation

Custom Object

Object Name	Support Ticket Intelligence
API Name	`Support_Ticket_Intelligence__c`

Fields

- Ticket Number
- Customer
- Contact
- Issue Type
- Description
- Priority Level
- Status
- Created Date
- Assigned To
- SLA Breach Risk
- Resolution Time

Working Process

Step	Action
Step 1	User creates a support ticket.
Step 2	The system retrieves the customer account.
Step 3	The latest support ticket is fetched.
Step 4	Agentforce reads the ticket description.
Step 5	The description is analyzed using keywords.
Step 6	Priority is predicted automatically.
Step 7	If the ticket is High Priority, an urgent task is created.
Step 8	The ticket is assigned to the appropriate support agent.
Step 9	A final response is generated.

Priority Detection Rules

High Priority

Keywords:
- urgent
- not working
- failure

Action:
- Create urgent task
- Assign to Senior Support Agent

Medium Priority

Keywords:
- issue
- slow
- delay

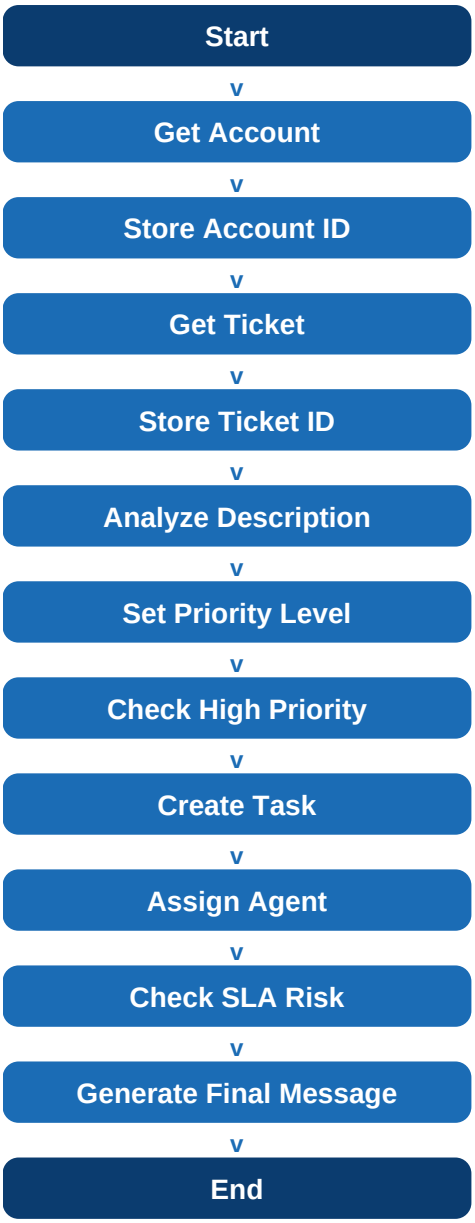
Action:
- Mark as Medium
- Queue for normal processing

Low Priority

If none of the above keywords are found.

Action:
- Mark as Low Priority
- Add to normal support queue

Flow Structure



Benefits

- Faster ticket handling
- Improved customer satisfaction
- Reduced manual effort
- Better workload management
- Increased productivity
- Intelligent support automation
- Improved response time
- Better SLA management

Expected Output

- Automatic priority prediction
- Automatic support agent assignment
- High-priority task creation
- Intelligent ticket processing
- Fast customer issue resolution

Future Enhancements

- Machine Learning based ticket prediction
- Email notification system
- SMS alerts
- Real-time dashboard
- Chatbot integration
- Predictive analytics
- Multi-language support
- Customer sentiment analysis
- Voice-based ticket creation
- Mobile application support

Sample Apex Code

```
public class TicketPriority {

    public static String getPriority(String description){

        description = description.toLowerCase();

        if(description.contains('urgent') ||
           description.contains('failure') ||
           description.contains('not working')){
            return 'High';
        }

        if(description.contains('issue') ||
           description.contains('slow') ||
           description.contains('delay')){
            return 'Medium';
        }

        return 'Low';
    }
}
```

Conclusion

The **Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce** is a smart Salesforce automation project that helps organizations classify support tickets based on urgency, automatically assign them to the appropriate support agent, and create urgent tasks for critical issues. By integrating Salesforce Flow with Agentforce, the system minimizes manual intervention, speeds up ticket resolution, improves customer satisfaction, and enhances overall support team productivity. It demonstrates how AI-powered automation can transform traditional customer support into an efficient, reliable, and intelligent service.